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Manuscript Title: Compact Cathode Architecture by Sustainable Natural Product-Derived Binder for High-Energy-Density Lithium–Sulfur Batteries

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We are pleased to submit our manuscript entitled *“Compact Cathode Architecture by Sustainable Natural Product-Derived Binder for High-Energy-Density Lithium–Sulfur Batteries”* for your consideration.

In this work, we present a comprehensive strategy to

…..

We believe our work delivers a compelling combination of electrochemical performance optimization and interfacial mechanical insight, contributing a valuable perspective to the field of metal-anode stabilization. We kindly ask that our manuscript be considered for publication in your esteemed journal.

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Sincerely,

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Compact Cathode Architecture by Sustainable Natural Product-Derived Binder for High-Energy-Density Lithium–Sulfur Batteries

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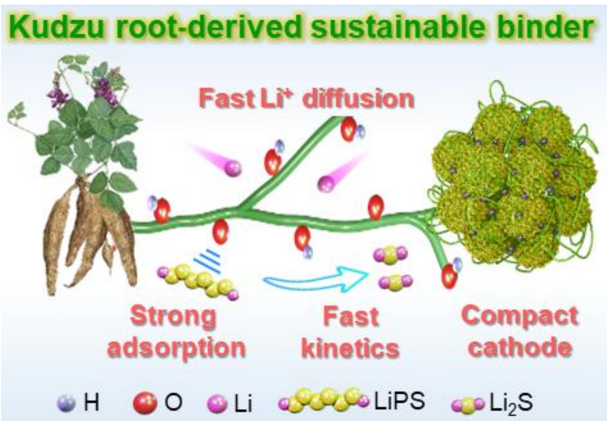
TOC graphic and summary:

"A sustainable binder derived from Kudzu root powder exhibits strong lithium polysulfide adsorption and catalytic activity, facilitates rapid lithium-ion diffusion, and regulates cathode architecture, thereby achieving high-energy-density lithium–sulfur batteries."



Abstract

The practical deployment of lithium–sulfur batteries is hindered by severe polysulfide shuttling, sluggish redox kinetics, and poor electrode structural stability. Binder engineering has emerged as a practical and effective strategy to address these challenges. Here, it is demonstrated that a sustainable, amphiphilic binder derived from Kudzu root powder simultaneously regulates lithium polysulfides, enhances charge transport, and enables electrode densification. The abundant hydroxyl groups in the binder provide strong polar anchoring sites for polysulfide adsorption, accelerate their redox conversion, and facilitate rapid lithium-ion transport. In parallel, the amphiphilic microstructure effectively regulates cathode architecture, enabling the formation of dense and mechanically robust electrodes with high sulfur loading. As a result, lithium–sulfur cells incorporating this binder deliver a high initial discharge capacity of 1214 mAh g^{-1} at 0.1 C and maintain 75% capacity retention over 500 cycles at 1C. Remarkably, a high cell-level volumetric energy density of 827 Wh L^{-1} is achieved under practical conditions, including a high sulfur loading of 11.3 mg cm^{-2}, a lean electrolyte of 5.0 μL mg^{-1}, a low negative-to-positive capacity ratio of 1.3, and an ultralow binder dose of 3%. This work highlights Kudzu root powder as a viable, sustainable binder for designing high-performance lithium–sulfur batteries and provides a promising pathway toward green energy storage systems based on renewable biomass resources.

Keywords: lithium–sulfur battery, binder, electrode architecture, energy density, natural product

Introduction

Lithium–sulfur (Li–S) batteries have emerged as one of the most compelling candidates for next-generation energy storage, driven by their ultrahigh theoretical energy density of 2600 Wh kg^{-1} or 2800 Wh L^{-1} and the natural abundance, low cost, and environmental benignity of elemental S *(1, 2)*. Despite these intrinsic merits, the practical deployment of Li–S technology remains severely hindered by several interrelated challenges: the electrically insulating nature of S and its discharge products (Li_2S_2/Li_2S), the dissolution and shuttling of intermediate lithium polysulfides (LiPSs, Li_2S_n, n = 4 – 8), sluggish solid–liquid–solid phase-conversion kinetics, and the large volumetric expansion (~80%) of the S cathode upon lithiation *(3, 4)*. What is worse, the demonstrated volumetric energy density of Li–S batteries is much behind the theoretical value, likely due to the low tap density of the S cathode (less than 1.0 g cm^{-3}) *(5, 6)*. These problems collectively manifest as rapid capacity fading, elevated polarization, and poor rate capability, causing a critical bottleneck between laboratory promise and commercial viability.

To address these challenges, extensive efforts have focused on host material and architecture engineering *(7–9)*, electrolyte formulation *(10)*, functional separator design *(11)*, and the development of advanced polymeric binders *(12)*. Among these strategies, binder engineering is increasingly recognized as one of the most practical and scalable strategies *(13)*. The binder serves as the structural backbone of the S composite cathode, governing particle-level adhesion among the active material, conductive carbon, and current collector *(12, 14)*. It also dictates the spatial distribution of S during slurry casting and drying; controls electrolyte wetting and ion-transport pathways throughout the electrode; and mediates electronic percolation across the composite framework *(15)*. Critically, in a system subject to large volumetric fluctuations and severe LiPS dissolution, the binder simultaneously must provide mechanical robustness, ionic accessibility, and chemical affinity for LiPS intermediates *(16)*. Therefore, to fulfill this combination of requirements, a judicious selection of a binder is vital to solving the inherent problems in Li–S batteries.

Conventional binders such as polyvinylidene fluoride (PVDF) suffer from poor ionic conductivity, negligible LiPS adsorption capability, and reliance on toxic N-methyl-2-pyrrolidone (NMP) as a processing solvent, motivating exploration of water-processable alternatives (*17*, *18*). Biomacromolecular binders derived from cellulose, alginate, and natural gums have attracted considerable attention in this context, offering high viscosity, robust mechanical strength, and abundant polar functional groups (–OH, –COOH) that help maintain electrode structural integrity and partially suppress the shuttle effect through electrostatic and hydrogen-bonding interactions with LiPS species. However, these binders require complex synthesis processes due to the incorporation of multiple components. Moreover, they are likely unable to effectively regulate cathode architecture to improve the tap density of S cathodes, particularly at high S loadings, thereby falling short of the multifunctional performance required for high-energy-density Li–S batteries.

Here, we report a sustainable, amphiphilic binder derived from Kudzu root powder that simultaneously addresses LiPS management, charge transport, and electrode densification in LSBs. The rich hydroxyl (–OH) functionality of the Kudzu binder provides strong polar anchoring sites for LSB adsorption, accelerates their redox conversion, and facilitates rapid Li-ion diffusion. Importantly, the amphiphilic microstructure of the Kudzu binder regulates the cathode architecture, enabling the construction of dense, mechanically robust cathodes with high active-material loading. Benefiting from synergistic effect, Li–S cells employing the Kudzu binder deliver an initial discharge capacity of 1214 mAh g^{-1} at 0.1C and maintain 75% capacity retention after 500 cycles at 1C. Notably, the S cathode still operated well under practical conditions, including a high S loading of 11.3 mg cm^{-2}, a low electrolyte-to-sulfur (E/S) ratio of 5.0 μL mg^{-1}, a low negative-to-positive (N/P) capacity ratio of 1.3, and even a reduced binder content of 3%.

Results and Discussion

Hypothesis for Multifunctional Binders

Kudzu root is an abundant, renewable, polysaccharide-rich plant that has long been used in traditional food and medicinal applications (*19*), yet its value as a functional materials resource remains underexplored. The starch extracted from Kudzu root is rich in hydroxyl-functionalized polysaccharides, which provide strong adhesive capability and robust intermolecular interactions. Inspired by the historical and modern use of natural starch-based materials, we envisioned that Kudzu-derived starch could serve as a sustainable and multifunctional binder for Li–S batteries.

Kudzu roots were processed through simple aqueous extraction to yield a fine white starch powder (Fig. 1a), which is a procedure requiring no organic solvents, no elevated temperatures, and no specialized reagents. Notably, Kudzu-derived starch is priced at just $41 per 500 g, approximately 19-fold lower than PVDF ($782 per 500 g) (Fig. 1b), positioning it as a compelling low-cost alternative to the fluorinated synthetic binders that currently dominate Li–S cathode fabrication and demand toxic NMP as a processing solvent.

The resulting polysaccharide was found to carry a dense array of hydroxyl (–OH) groups along its amylose and amylopectin backbone, and we expected that these polar functional groups would simultaneously provide strong electrode adhesion, chemical anchoring of LiPS intermediates, and enhanced electrolyte wettability. When the Kudzu binder was mixed with S, CNTs, and Super P, we anticipated that the electrode components would be integrated into densely packed, uniform agglomerates with strong cohesion, a structural outcome considered unachievable through the passive physical attachment offered by PVDF alone (Fig. 1c).

We further hypothesized that soluble LiPS species would be immobilized at the electrode surface through cooperative hydrogen-bonding interactions with the –OH-rich backbone, thereby suppressing shuttle-driven capacity fade. Simultaneously, it was anticipated that the CNT scaffold, held in intimate contact by the Kudzu matrix, would establish continuous electron pathways throughout the cathode interior, and we expected that rapid Li^+ diffusion would be promoted by the hydrophilic polysaccharide environment, collectively accelerating sulfur redox kinetics (Fig. 1d). On this basis, we hypothesized that the Kudzu binder would simultaneously fulfil the roles of structural adhesive, polysulfide regulator, and ion/electron transport facilitator, functions that are addressed in conventional binder systems only through multiple specialised components and at considerably greater materials cost. The electrochemical validation of these hypotheses is presented in the sections that follow.

Characterizations of Kudzu Binder

Initially, Kudzu root powder is insoluble in water. However, upon heating, it undergoes gelatinization to form an homogenious solution (Fig. 2a), which can be directly used as a binder in cathode slurry preparation. This water-processable feature renders the KRP binder more sustainable than conventional PVDF, which requires the toxic organic solvent NMP for dissolution. After drying, white foam is obtained (Fig. S1).

The morphological evolution of Kudzu root powder was investigated using a scanning electron microscope (SEM). The pristine KRP has a spherical morphology with particle diameters below 10 μm (Fig. 2b and Fig. S2). After gelatinization and drying, a three-dimensional interconnected network is generated (Fig. 2c), which facilitates structural integrity and electronic/ionic connectivity throughout the sulfur cathode.

The crystallographic structure and chemical composition were further characterized by X-ray diffraction (XRD), Fourier transform infrared spectroscopy (FT-IR), and X-ray photoelectron spectroscopy (XPS). The XRD pattern of raw KRP displays distinct crystalline peaks at $2\theta \approx 15.3°$, $17.3°$, and $22.8°$ (Fig. 2d) *(20)*. Notably, these peaks disappear after gelatinization, indicating disruption of the ordered structure and transformation into an amorphous phase. Such amorphization, similar to other polymer matrices with enhanced segmental mobility and increased free volume, is beneficial for Li^+ transport *(21)*. The FI-IR spectra of KRP before and after gelatinization show similar characteristic features (Fig. 2e). The peak near 993 cm^{-1} is assigned to C–O–C asymmetric stretching, while the bands at ~1081 and 1153 cm^{-1} correspond to C–O–H stretching vibrations *(20)*. The presence of abundant C–O functionalities is further confirmed by XPS (Fig. 2f), where a strong peak at

286.0 eV is observed. These C–O–C and C–O–H groups are expected to serve as active sites for LiPS adsorption and catalytic conversion, while also facilitating Li^+ transport within the electrode.

The chemical composition and glycosidic linkages of the Kudzu binder were elucidated through chemical derivatization combined with nuclear magnetic resonance (NMR) spectroscopy. Monosaccharides obtained by acid hydrolysis of the Kudzu binder were reacted with L-cysteine methyl ester hydrochloride to form thiazolidine derivatives. ^{1}H NMR data of these derivatives exhibited signals corresponding to D-glucose (δ_H 5.73 (d, J = 7.2 Hz) and δ_H 5.73 (d, J = 7.2 Hz) (Fig. S3) *(22)*, confirming that D-glucose is the building block of the Kudzu binder. Furthermore, the ^{1}H NMR characterization of the intact binder dissolved in DMSO-*d6* revealed that these glucose unit are connected through α-(1→4) and α-(1→6) glycosidic bonds (Fig. 2g) *(23)*. The liner chains formed by α-(1→4)-linkages are interconnected through α-(1→6) branching points, resulting in a highly branched macromolecular networks

Polysulfide Adsorption and Catalytic Conversion

To suppress the LiPS shuttle effect, the cathode materials should have a high adsorption capability to keep LiPSs at the cathode. They should also have high catalytic activity to accelerate conversion reactions *(24)*. To understand the interaction between the Kudzu binder and LiPSs at the atomic scale, density functional theory (DFT) calculations were performed. …

To validate the theoretical predictions, a series of experiments was conducted. Visual adsorption tests were first performed to qualitatively assess the adsorption capabilities of Kudzu and PVDF binders. Equal amounts of each binder were immersed in a Li_2S_6 solution. After 3 h, the solution containing the Kudzu binder became nearly colorless, whereas the PVDF-containing solution remained nearly unchanged (Fig. 3d and Fig. S4). Consistently, UV–vis measurements show a markedly diminished LiPS signal in the Kudzu-containing solution, indicating that most LiPS species were effectively captured (Fig. 3e). These results demonstrate the strong adsorption affinity of Kudzu toward LiPSs, which can effectively suppress their diffusion into the electrolyte (25). To further elucidate the interaction mechanisms, comparative XPS analysis was performed on Kudzu and PVDF binders after the adsorption test (Fig. 3f). In the case of PVDF, no detectable sulfur signal is observed, suggesting negligible chemical interaction with LiPSs. In sharp contrast, Kudzu exhibits a pronounced S 2p signal. Distinct peaks corresponding to terminal sulfur (S_T^{-1}) and bridging sulfur (S_B^0) confirm the anchoring of LiPSs on the Kudzu binder. Moreover, the prominent thiosulfate and polythionate signals further indicate that chemical interactions between the C–O–C and –OH functional groups of Kudzu and LiPSs likely occur (16). These results verified that the Kudzu binder has a strong ability to chemically anchor LiPSs.

The catalytic activity of the Kudzu binder was evaluated using both symmetric and asymmetric cell configurations. Symmetric cells were assembled with two identical electrodes composed of carbon nanotubes (CNTs), Super P, and either Kudzu or PVDF binder, with Li_2S_6 solution serving as the catholyte. In cyclic voltammetry (CV) measurements, the cell employing the Kudzu binder delivers higher current responses and exhibits more pronounced redox peaks compared to the PVDF-based cell (Fig. 3g). Notably, even at higher scan rates, well-defined redox peaks are maintained in the Kudzu cell (Fig. S5), confirming its superior catalytic activity. Moreover, the smaller semicircle at the high frequency region in the electrochemical impedance spectroscopy (EIS) analysis reveals the faster charge transfer in the Kudzu-based cell (Fig. S6). In asymmetric cells, sulfur cathodes with a loading of ~1.2 mg cm^{-2} were paired with Li metal anodes. The enhanced electrocatalytic performance of the Kudzu binder is evidenced by a negative shift in the anodic peaks and a positive shift in the cathodic peaks (Fig. 3h and Fig. S7), indicating reduced polarization during liquid–solid conversion and accelerated kinetics of S redox reactions (26). Additionally, the sharper and more intense peaks further suggest more efficient LiPS conversion and faster charge transfer. To quantitatively analyze the reaction kinetics, Tafel slopes were extracted from the linear regions of the redox peaks (Fig. 3i and Fig. S8). Compared with the PVDF-based cell, the Kudzu cell exhibits significantly lower Tafel slopes of 20 and 86 mV dec^{-1} for the cathodic and anodic processes, respectively, indicating faster charge transfer. These results demonstrate that the Kudzu binder effectively catalyzes LiPS conversion, leading to improved sulfur utilization.

Theoretically, ~75% of the capacity in Li–S batteries originates from the conversion of soluble Li_2S_4 intermediates to the insoluble discharge product Li_2S (27). However, this liquid–solid transformation involves a substantial energy barrier, resulting in high overpotential. Incorporating a catalytically active binder can lower this barrier and promote Li_2S formation. To evaluate this effect, potentiostatic discharge measurements were conducted to probe Li_2S nucleation behavior. The Kudzu-based cell exhibits a higher current response and faster nucleation kinetics compared to the PVDF-based cell, indicating facilitated Li_2S formation (Fig. 3j). SEM analysis further reveals that Li_2S deposition on the PVDF-based electrode is highly agglomerated, suggesting preferential nucleation at localized sites (Fig. 3k). In contrast, the Kudzu-based electrode shows uniform Li_2S nucleation, leading to dense and homogeneous deposition (Fig. 3l). Collectively, these results confirm that the Kudzu binder effectively catalyzes sulfur conversion in Li–S batteries.

Li-Ion Diffusion

The …

Cathode Architecture Regulation

The energy density of Li–S batteries strongly depends on the compactness of the S cathode (28). It has been demonstrated that S cathodes with high tap density can significantly enhance the overall energy density of Li–S batteries (29). As the structural backbone of the S composite cathode, the binder plays a critical role in determining the electrode architecture.

In this work, S cathodes composed of S, carbon nanotubes (CNTs) as the S host, Super P, and either PVDF or Kudzu as the binder were prepared. The SEM images of the PVDF-based cathode (Fig. 5a,b and Fig. S10) reveal a highly loose and porous architecture. Large cracks are distributed throughout the electrode, indicating insufficient interparticle cohesion and weak structural integrity (17). At higher magnification, the PVDF binder appears unable to uniformly integrate S particles and conductive additives into a compact framework, resulting in discontinuous contact between components. Consequently, at an S mass loading of 7.1 mg cm^{-2}, the electrode exhibits a large thickness of 118 μm (Fig. 5c). Such a thick and porous structure leads to a low tap density of only 0.94 mg cm^{-3}, implying substantial internal void space within the cathode. These pores must be filled with electrolyte during cell operation, thereby increasing electrolyte consumption and reducing both the gravimetric and volumetric energy density of the battery (6, 7). In addition, the presence of cracks may interrupt electron transport pathways and accelerate structural degradation during cycling.

In contrast, the cathode employing the Kudzu binder (Fig. 5d,e) exhibits a markedly different morphology. The S composite forms densely packed spherical agglomerates that are tightly interconnected, generating a continuous conductive framework throughout the electrode (Fig. S10). Unlike the fractured morphology observed for PVDF, the Kudzu-based electrode is essentially crack-free, indicating significantly improved mechanical cohesion among S, CNTs, and conductive carbon particles. Once immersed in the electrolyte, the electrodes can remain stable for an extended period without noticeable detachment (Fig. S11). The interconnected spherical architecture (Fig. S12) is expected to promote efficient electron transport while simultaneously maintaining structural stability during electrochemical cycling. This compact organization dramatically reduces the electrode thickness to 91 μm. Correspondingly, the tap density increases from 0.94 to 1.25 mg cm^{-3}, representing a ~33% enhancement in electrode compactness. The higher tap density suggests that the Kudzu-based cathode contains substantially fewer internal voids and therefore requires less electrolyte infiltration. Reducing excess electrolyte demand is highly beneficial for practical Li–S batteries because it improves sulfur utilization under lean-electrolyte conditions and increases the overall cell-level energy density while lowering manufacturing cost (30).

Additional control experiments provide insight into the origin of these spherical structures. Spherical agglomerates were observed only when S was present, but not in composites containing only CNTs or Super P (Fig. S13). This strongly suggests that the Kudzu binder directly interacts with S species rather than with the conductive carbon additives. Furthermore, similar agglomerate formation was also observed when reduced graphene oxide was used as the S host (Fig. S14), indicating that the phenomenon is independent of the type of carbonaceous matrix. Therefore, the compact electrode architecture primarily originates from a specific Kudzu–S interaction.

Overall, these results indicate that the conventional PVDF binder is unable to effectively integrate S and conductive components into a compact and mechanically stable electrode structure (Fig. 5g). In contrast, the Kudzu binder enables the formation of densely packed, crack-free, interconnected spherical agglomerates, thereby significantly improving electrode compactness, tap density, and potentially the practical energy density of Li–S batteries (Fig. 5h).

Electrochemical Performance

The superiority of the Kudzu binder over conventional PVDF was further validated in full-cell Li–S batteries. The sulfur cathodes were prepared with a sulfur loading of 1.5 mg cm^{-2} and a binder content of 10 wt%. As shown in Fig. 6a, the cell employing the Kudzu binder delivered a high initial discharge capacity of 1214 mAh g^{-1} at 0.1 C, which was substantially higher than that of the PVDF-based cell (918 mAh g^{-1}). In addition, the galvanostatic charge–discharge (GCD) curves of the Kudzu cell exhibited smaller voltage polarization and reduced overpotentials for Li_2S oxidation/reduction processes (Fig. S16), indicating accelerated sulfur redox kinetics induced by the Kudzu binder.

The EIS analysis further revealed that the Kudzu cell possessed the lowest charge-transfer resistance, as evidenced by the smallest semicircle diameter in the high-frequency region (Fig. S17). Meanwhile, owing to the strong affinity of the Kudzu binder toward LiPSs, the corresponding cell exhibited a significantly suppressed shuttle current (Fig. S18), confirming the effective confinement of soluble intermediates.

The Kudzu-based cell also demonstrated excellent rate capability. The discharge capacities at current densities of 0.1, 0.2, 0.5, 1, 2, and 4 C reached 1214, 1086, 997, 881, 745, and 571 mAh g^{-1}, respectively (Fig. 6b). Notably, when the current density was returned to 0.1 C, a high reversible capacity of 1192 mAh g^{-1} was recovered, highlighting the excellent reversibility and structural robustness of the sulfur cathode. The superior rate performance can be attributed to the synergistic effects of strong LiPS adsorption, enhanced catalytic activity, and rapid ion transport enabled by the Kudzu binder. As presented in Fig. 6c and Fig. S19, the characteristic charge/discharge plateaus were still well maintained even at high current densities, indicating fast and efficient LiPS conversion reactions.

More importantly, the multifunctional characteristics of the Kudzu binder resulted in remarkable long-term cycling stability. At 0.1 C, the Kudzu cell delivered an initial capacity of 1232 mAh g^{-1} and retained 81% of its capacity after 100 cycles (Fig. 6d). By contrast, the PVDF-based cell suffered from more severe capacity fading and retained only 72% of its initial capacity over the same cycling period. Even at a high current density of 1 C, the Kudzu cell maintained exceptional cycling durability with a low capacity fading rate of only 0.038% per cycle. After 500 cycles, the cell still delivered a high reversible capacity of 804 mAh g^{-1} (Fig. 6e), corresponding to a significantly smaller decay rate than that of the PVDF cell (0.053% per cycle).

After long-term cycling tests, the coin cells were disassembled for further analysis. The separator and Li anode from the Kudzu-based cell exhibited a light-yellow coloration, further confirming the excellent LiPS adsorption capability of the Kudzu binder (Fig. S20). Moreover, the S cathode employing the Kudzu binder maintained its structural integrity, featuring dense Li_2S deposition and minimal volume expansion.

The practical applicability of the Kudzu binder was further evaluated under demanding conditions using a high sulfur loading of 7.1 mg cm^{-2} and a lean electrolyte condition of 5 μL mg^{-1}. While the PVDF-based cell exhibited rapid capacity decay during the initial cycles, the Kudzu-based cell maintained stable cycling behavior over extended operation. The cell delivered an initial areal capacity of 6.54 mAh cm^{-2} and still retained 73% after 70 cycles at 0.1 C (Fig. 6f). The gravimetric and volumetric energy densities at the cell level were estimated to be 827 Wh L^{-1} and 273 Wh kg^{-1}, respectively (Table S1). These values are believed to increase further upon improving the adsorption capability and catalytic activity of the sulfur host. Remarkably, stable cycling performance could still be achieved even when the Kudzu binder content was reduced to an ultralow level of 3 wt% (Fig. 6g and Fig. S21), further highlighting its strong binding capability and multifunctional role in practical Li–S batteries. Overall, the performance of the Li–S cell employing the Kudzu binder is comparable to, or even surpasses, that of previously reported systems, highlighting the effectiveness of the Kudzu binder in enabling high-performance Li–S batteries (Table S2).

Conclusions

In summary, we demonstrated a sustainable binder derived from Kudzu root powder that couples polysulfide regulation with electrode densification. Its abundant hydroxyl groups provide polar anchoring sites that suppress polysulfide shuttling, accelerate redox conversion, and facilitate rapid lithium-ion transport, while its amphiphilic microstructure regulates the cathode architecture to yield dense, mechanically robust electrodes at high sulfur loading. Cells using this binder delivered 1214 mAh g^{-1} at 0.1 C and retained 75% of their capacity over 500 cycles at 1 C, reaching a cell-level volumetric energy density of 827 Wh L^{-1} under practical conditions (11.3 mg cm^{-2} sulfur, 5.0 μL mg^{-1} electrolyte, N/P ratio of 1.3, and 3 wt% binder). These results establish Kudzu root powder as a low-cost, high-performance binder for Li–S batteries, highlighting a viable pathway toward green, next-generation energy-storage systems based on renewable biomass resources.

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Conflict of Interest

The authors declare no conflicts of interest.

Supporting Information

The Supporting Information is available at …

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Figures

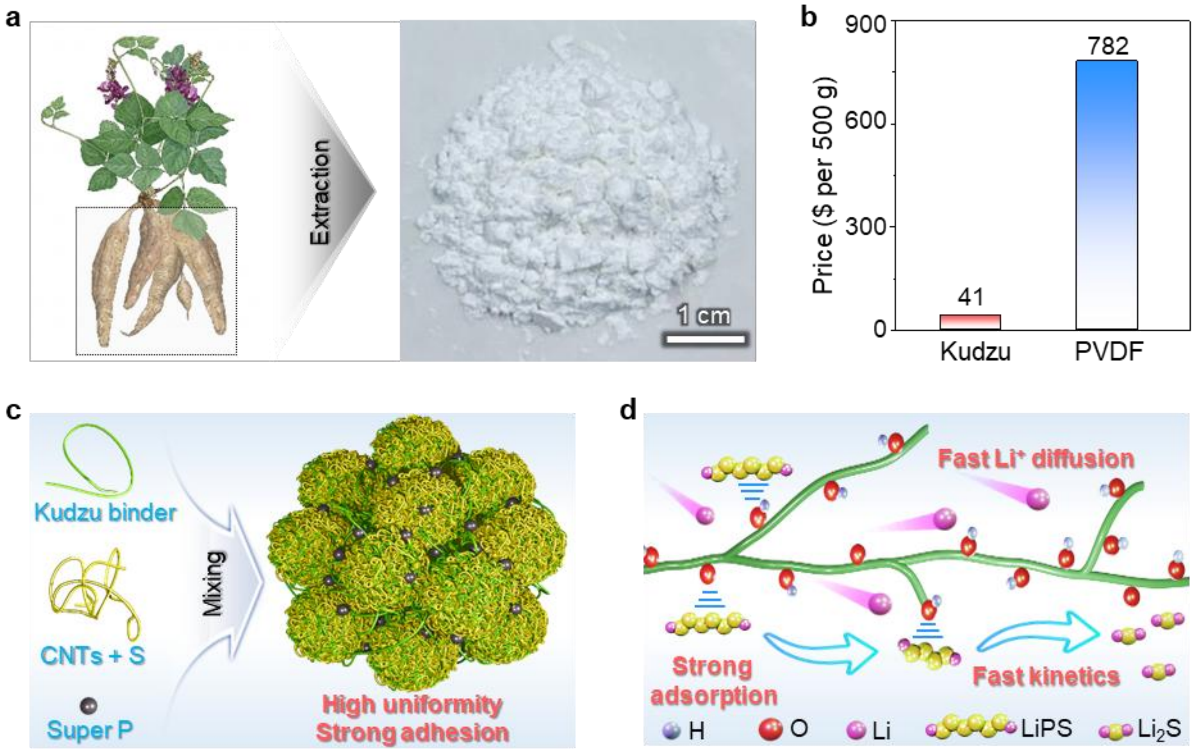

Figure 1. Kudzu root powder as a sustainable binder. a, Schematic illustration showing the origin and appearance of the Kudzu binder. **b,** Price comparison between the PVDF and Kudzu binder. **c,** Construction of S cathodes using the Kudzu binder. **d,** Expected effects created by the Kudzu binder.

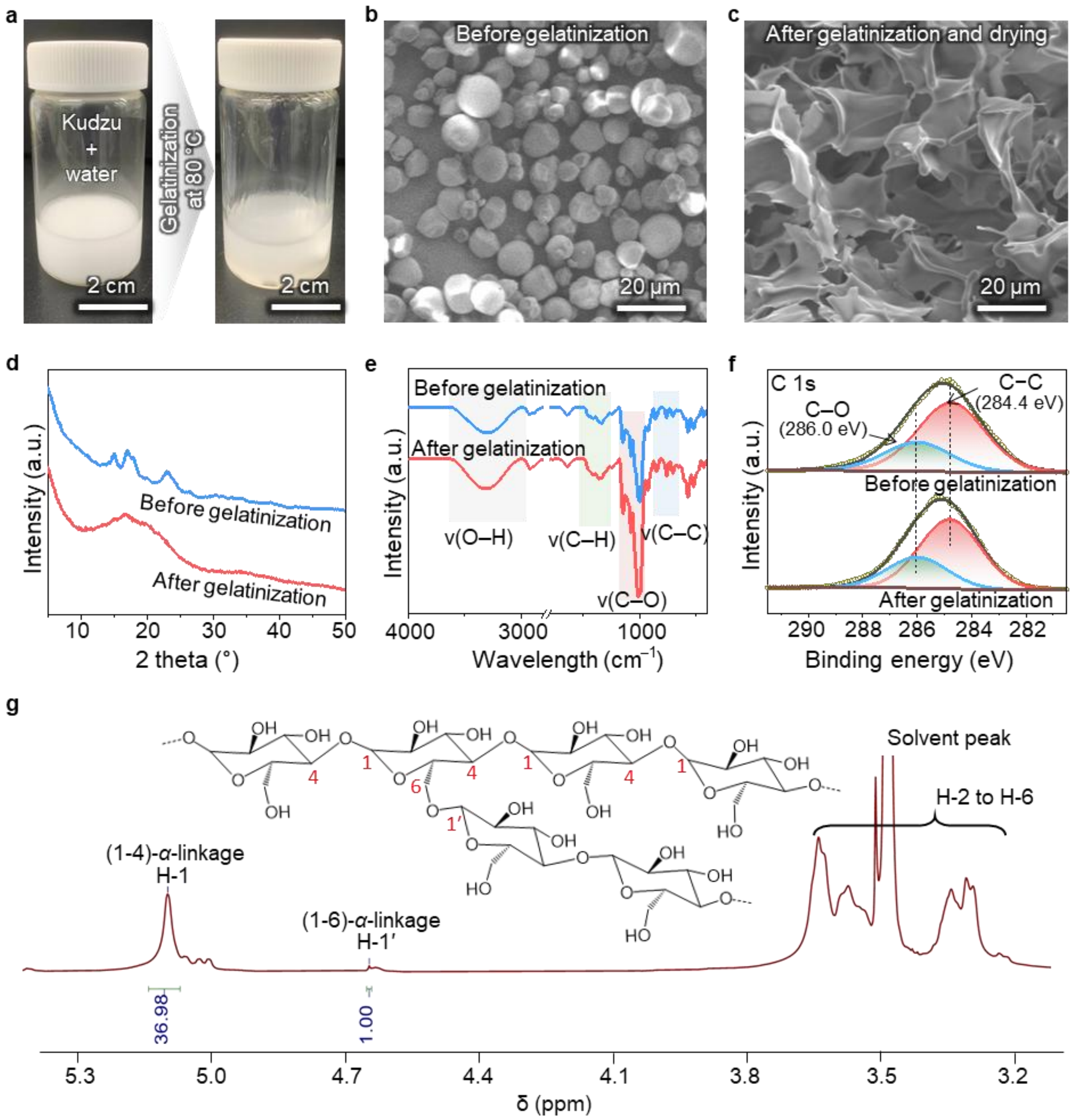

Figure 2. Characterization of the Kudzu binder. a, Preparation of the Kudzu binder solution. **b,c,** SEM images showing the morphologies of the Kudzu binder before (b) and after (c) gelatinization, respectively. **d–f,** XRD patterns (d), FT-IR spectra (e), and high-resolution C 1s XPS spectra (f) showing the structure and chemical bonding evaluation of the Kudzu binder before and after gelatinization. **g,** MNR spectra and relative configuration of the Kudzu binder after gelatinization.

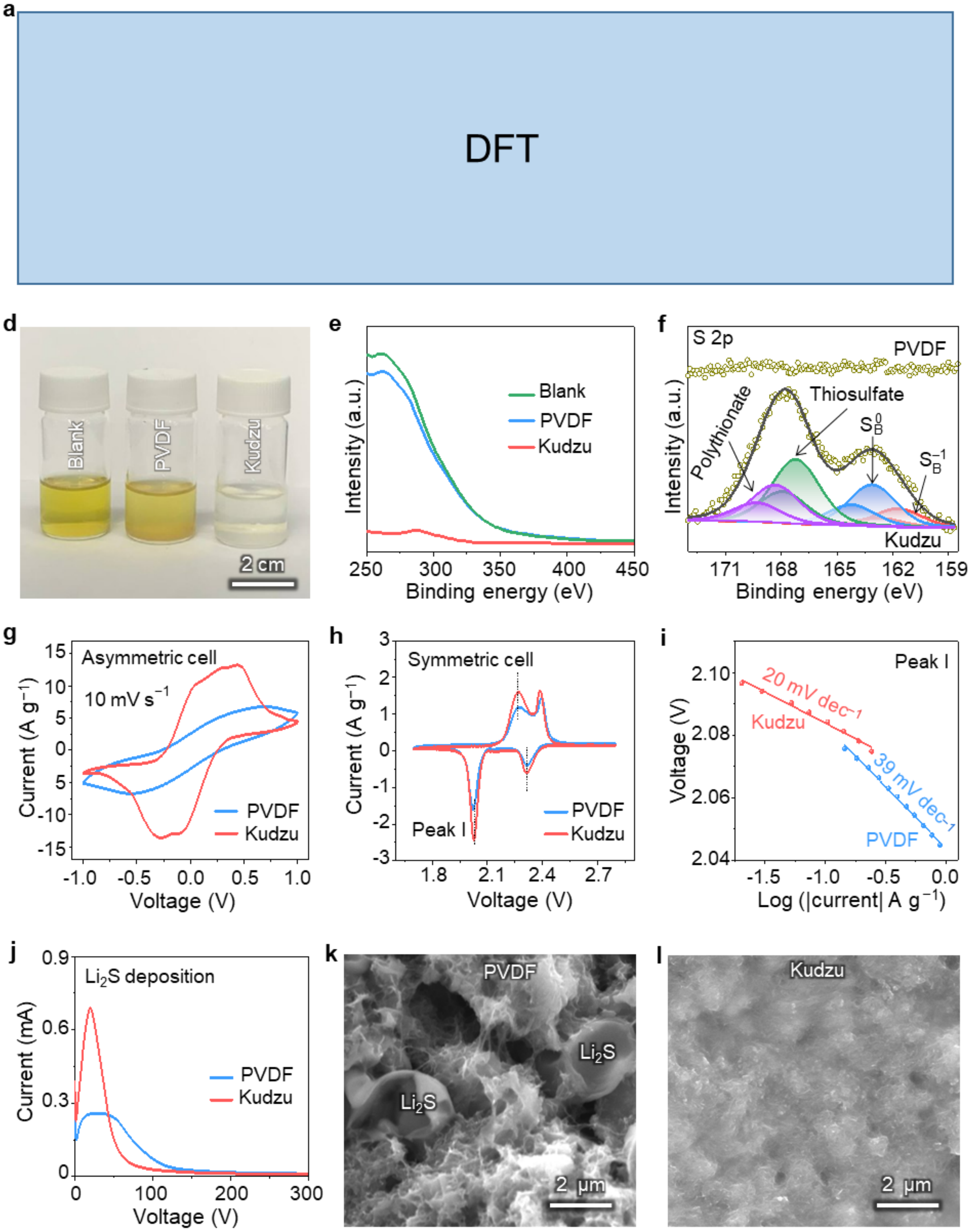

Figure 3. Interaction between the binders and LiPSs. a–c, DFT calculations. **d,** Optical photographs of visual Li2S6 adsorption test. **e,** UV–vis absorbance spectrum of Li_2S_6 solution after adsorption test. **c,** High-resolution XPS spectrum of S 2p in the PVDF and Kudzu binders after adsorption test. **j,** Potentiostatic discharge curves of Li_2S_8 electrolyte at 2.05 V for

evaluating the Li_2S nucleation. **k, l,** Morphologies of Li_2S nucleation on the S cathodes using the PVDF (k) and Kudzu (l) binders.

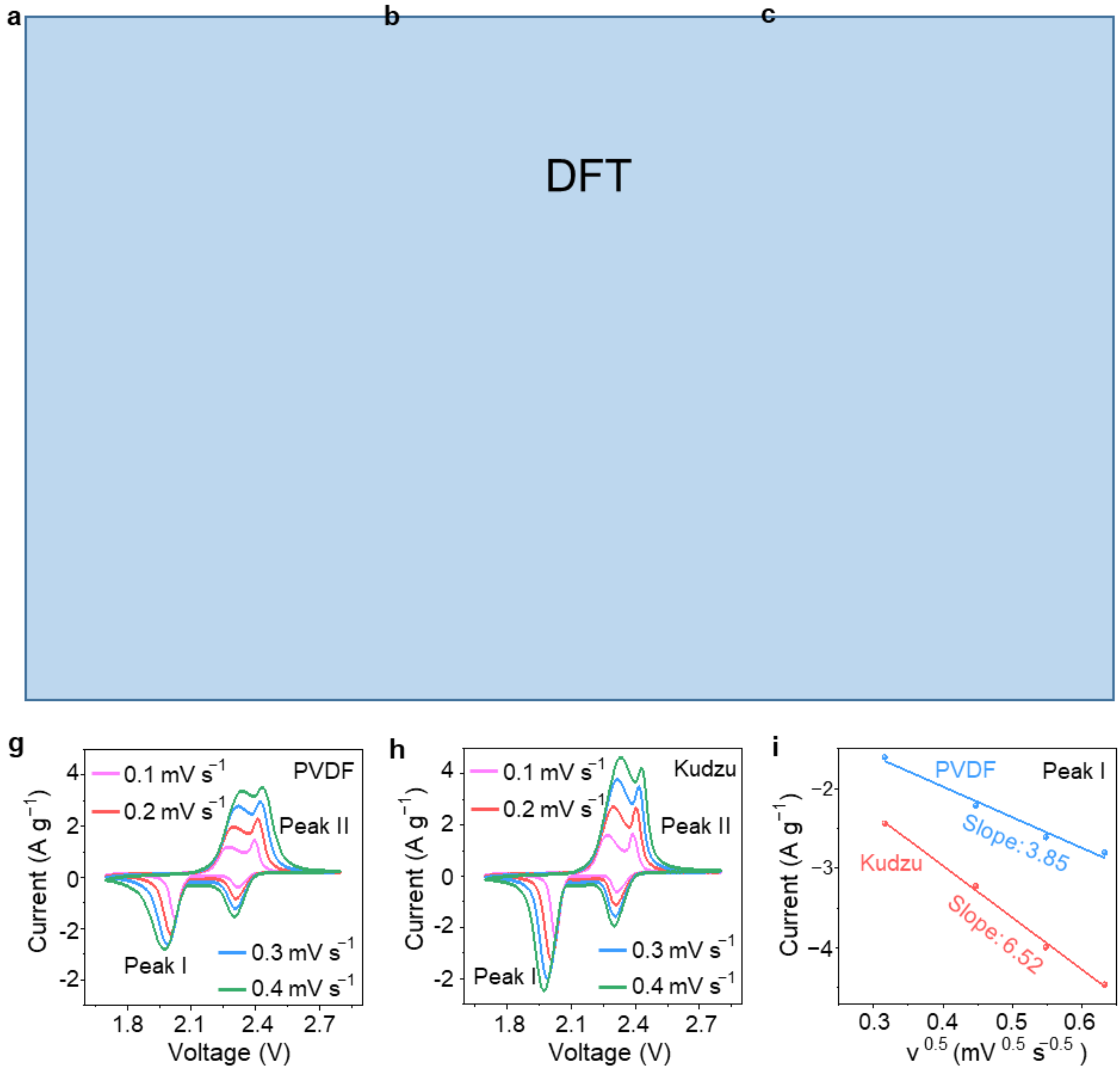

Figure 4. Li-ion diffusion characterization. a–c, DFT calculation. **g, h,** CV curves of the Li–S cells using the PVDF (g) and Kudzu (h) binders at various scan rates. **i,** Linear fitting of peak current versus the square root of the scan rates for peak I.

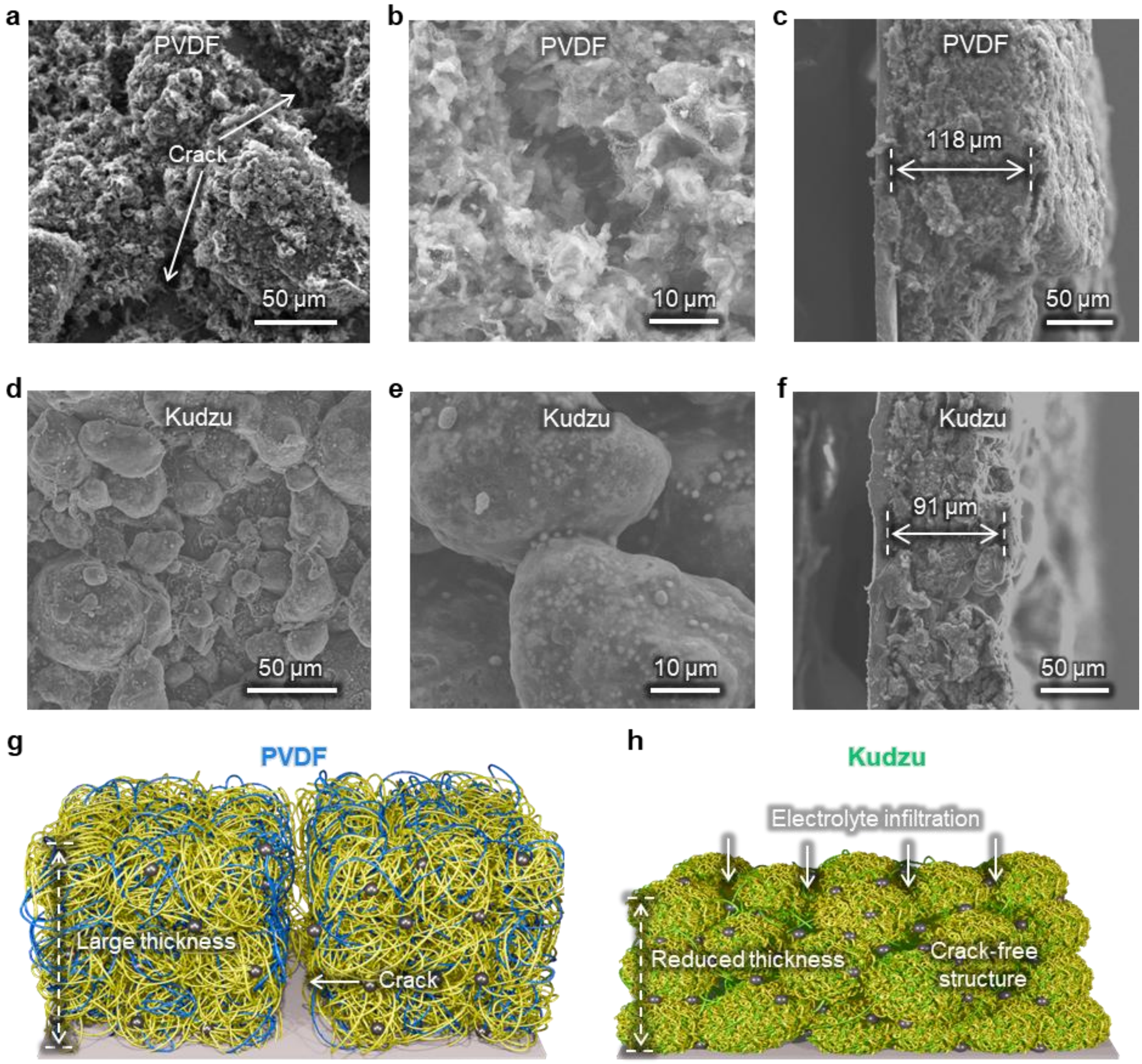

Figure 5. Morphological characterization of S cathodes. a,b, Top-view SEM images of the S cathode using PVDF at low and high resolution, respectively. **c,** Cross-section SEM images of the S cathode using PVDF. **d,e,** Top-view SEM images of the S cathode using the Kudzu binder at low and high resolution, respectively. **f,** Cross-section SEM images of the S cathode using the Kudzu binder. **g,h,**

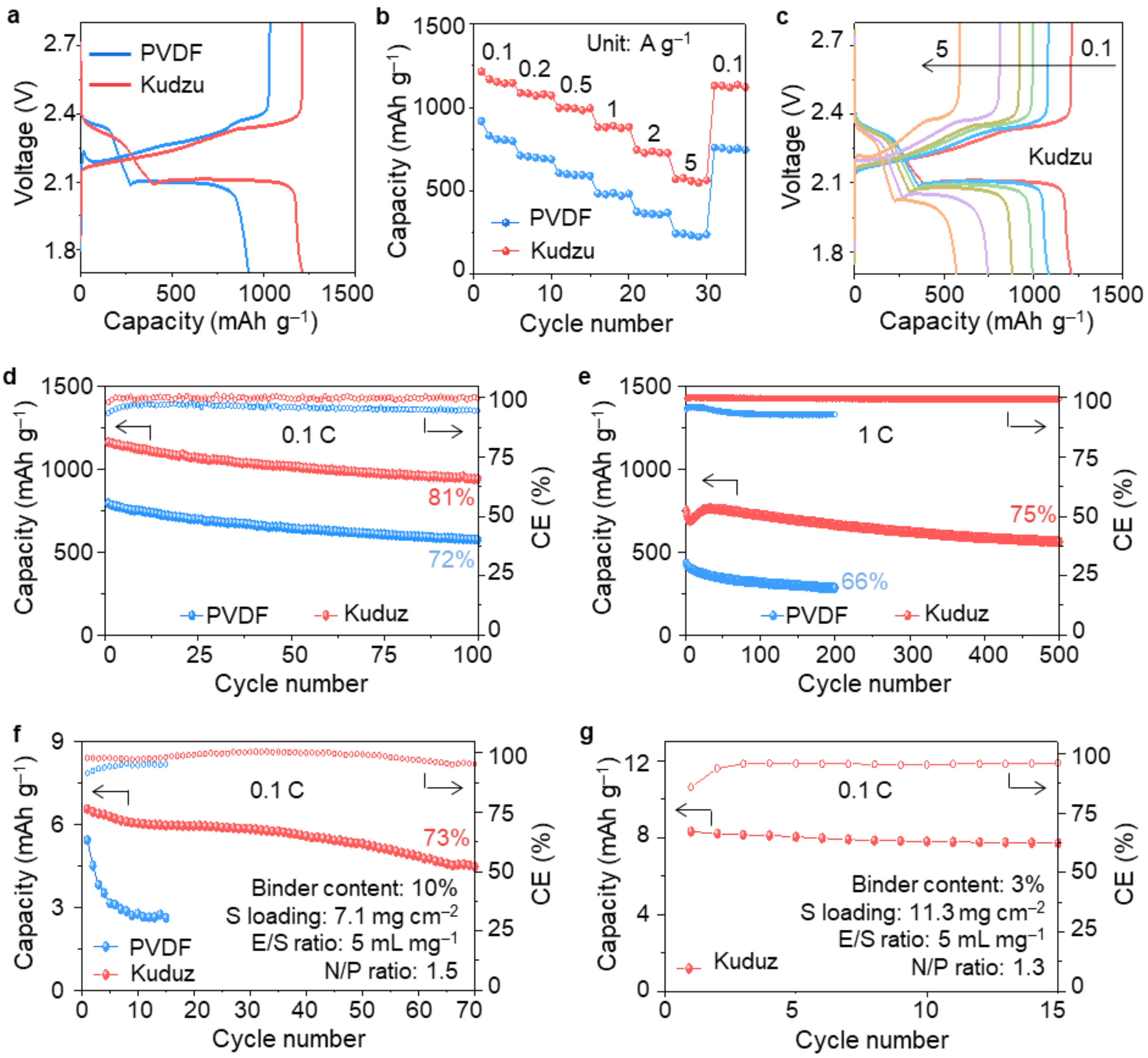

Figure 6. Electrochemical performance of LSB cells using the PVDF- and Kudzu-based S cathodes. a, Representative charge–discharge curves at 0.1 C. **b,** Rate capability. **c,** Representative charge–discharge curves of the LSB cells using the Kudzu-based S cathodes at various current densities. **d,e,** Cycling performance at 0.1 and 1 C. **f,** Cycling performance of the practical cell using extremely high S loading at 0.1 C. **g,** Representative charge–discharge curves of the practical cell using extremely high S loading and low content of the Kudzu binder.

SUPPORTING INFORMATION

Compact Cathode Architecture by Sustainable Natural Product-Derived Binder for High-Energy-Density Lithium–Sulfur Batteries

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1. Experimental Section

Preparation of Kudzu Root Powder

Fresh Kudzu roots (Pueraria lobata) collected in Hung Yen, Viet Nam were cleaned, peeled, and cut into small pieces. The root pieces were blended with deionized water and filtered through cheesecloth. The filtrate was allowed to stand overnight to sediment the starch. The supernatant was discarded, and the starch precipitate was washed several times. The collected starch was dried at 50 °C under vacuum, ground into a fine powder, and used as the Kudzu binder without any modification.

Material Characterizations

The analysis of the morphology and the elements were carried out using FE-SEM (JEOL, JSM-7800F) and EDX (Oxford AZtec® EDX system), respectively. XRD spectrum was collected using an Empyrean diffractometer (PANalytical with Cu Kα (λ = 1.5418 Å)). MultiLab

2000 (Thermo Fisher Scientific with an Al Kα X-ray source) was employed to obtain XPS spectra. A V-770 UV-VIS/NIR spectrophotometer was used to collect UV-visible absorption spectra of Li_2S_6 solution after polysulfide adsorption tests.

Structure Elucidation

Monosaccharide determination:

Kudzu powder was hydrolyzed with 0.3N HCl at 90 °C. The hydrolysis process was monitored by Thin-Layer Chromatography (TLC) at 0, 3, 6, 8, 10, 12, 15 h. Pure D-glucose purchased from Junsei (Japan) was used as the reference standard. After hydrolysis, the monosaccharide moieties were confirmed to be D-glucose via thiazolidine derivatives and NMR analysis, following the reported method (S1).

Linkage determination:

Kudzu powder (5.0 mg) was first gelatinized, followed by dehydration and dissolution in 500 µL deuterated DMSO for NMR measurement. NMR condition: 11 µs 90° pulse, a repetition time of 15.07 s composed of an acquisition time of 3.07 s and a relaxation delay of 12 s, 128 scans.

LiPS Adsorption Tests

0.5 M Li_2S_6 solution was prepared by mixing Li_2S and sulfur in a molar ratio of 1:5 in DOL/DME solution (1:1 by volume), followed by stirring vigorously at 70 °C for 24 h. PVDF and Kudzu root powder with the same weight were added in sealed vials that contain as-prepared Li2S6 solution (10 µL) diluted in another DOL/DME solution (2 mL). The digital photos to estimate the adsorption efficiency of the samples for lithium polysulfides were taken after 5h.

Li_2S Nucleation and Dissolution Measurements

CNTs were mixed with super P and PVDF or Kudzu binder and coated on Al foil, yielding electrodes with an identical CNT loading of 1 mg cm^{-2}. Lithium metal foil and Celgard 2500 membranes were employed as the anode and separator, respectively. To assemble the cells, 20 μL of 0.25 M Li_2S_8 catholyte was deposited onto the cathode, while 20 μL of blank electrolyte was added to the lithium anode side. The Li_2S_8 catholyte was prepared by dissolving Li_2S and S in a molar ratio of 1:7 in 1 M LiTFSI electrolyte at 70 °C for 24 h.

For the Li_2S nucleation test, the cells were first galvanostatically discharged to 2.06 V at a current of 0.112 mA, followed by a potentiostatic discharge at 2.05 V until the current decreased below 10^{-5} A, enabling Li_2S nucleation. The cells were subsequently discharged to 1.70 V at 0.10 mA to ensure complete conversion of soluble lithium polysulfides into solid Li_2S on the catalyst surface. The resulting Li_2S was then oxidized through galvanostatic charging to 1.80 V at 0.01 mA, followed by potentiostatic charging at 2.35 V until the current dropped below 10^{-5} A to evaluate Li_2S dissolution behavior.

Li–S Cell Assembly and Electrochemical Characterization

S was infiltrated in CNTs at 156 °C for 12 h with the S content of 80%. The composite of S/CNTs and super P was uniformly mixed with PVDF in NMP or Kudzu binder in DI water. The mass ratio among S/CNTs, super P, and binder was 8:1:1. The resulting slurry was cast onto aluminum foil and dried under vacuum at 50 °C for 24 h.

The electrolyte consisted of 1 M lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) dissolved in a 1:1 (v/v) mixture of 1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME), with 1 wt.% $LiNO_3$ as an additive. Lithium metal foil (100 μm thick) and Celgard 2500 membranes were used as the anode and separator, respectively.

For cells with a sulfur loading of 1.5 mg cm^{-2}, the electrolyte-to-sulfur (E/S) ratio was maintained at 10 μL mg^{-1}. When the sulfur loading was increased to 7.1 mg cm^{-2}, the E/S ratio of 5.0 μL mg^{-1} and 50-μm Li disks were employed.

Cyclic voltammetry (CV) measurements were performed on a VMP3 electrochemical workstation over a voltage range of 1.7–2.8 V (vs. Li^+/Li) at a scan rate of 0.1 mV s^{-1}. Electrochemical impedance spectroscopy (EIS) measurements were conducted on the same workstation within a frequency range of 100 kHz to 0.01 Hz. Galvanostatic charge–discharge tests were carried out using a WonATech battery cycler within a voltage window of 1.7–2.8 V at room temperature. Prior to long-term cycling, all cells were activated for three cycles at 0.1 C and subsequently cycled at 1.0 C. Here, 1 C corresponds to a theoretical current density of 1,675 mA g^{-1} based on the sulfur mass.

2. Supporting Figures

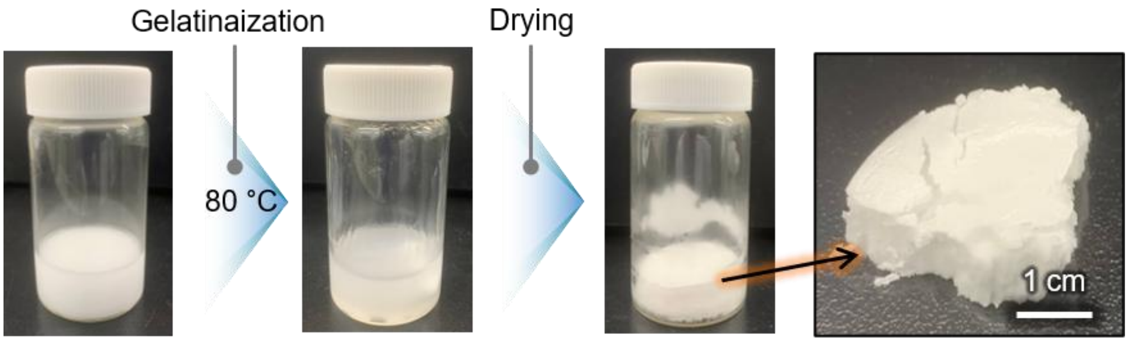


Figure S1. Enlarged SEM images showing morphologies of (A) bare Zn and (B) ZnF_2(90)/Zn.

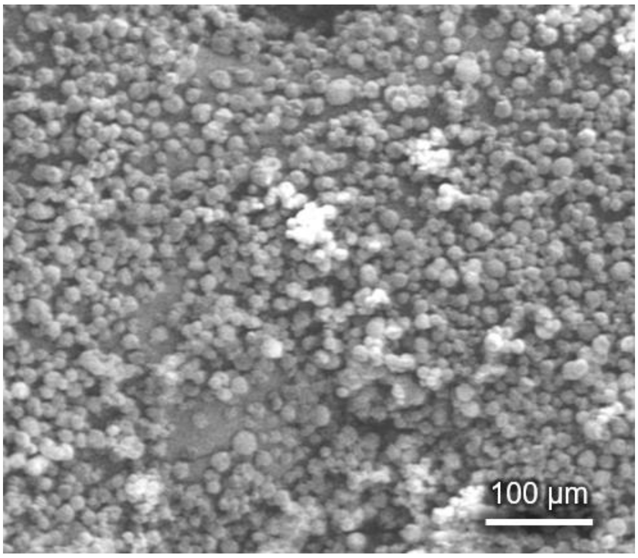
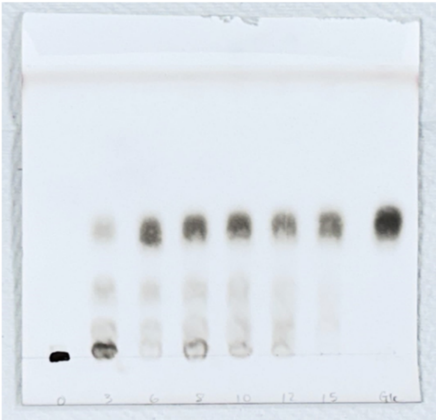


Figure S2. SEM image of the Kudzu powder.

a



b

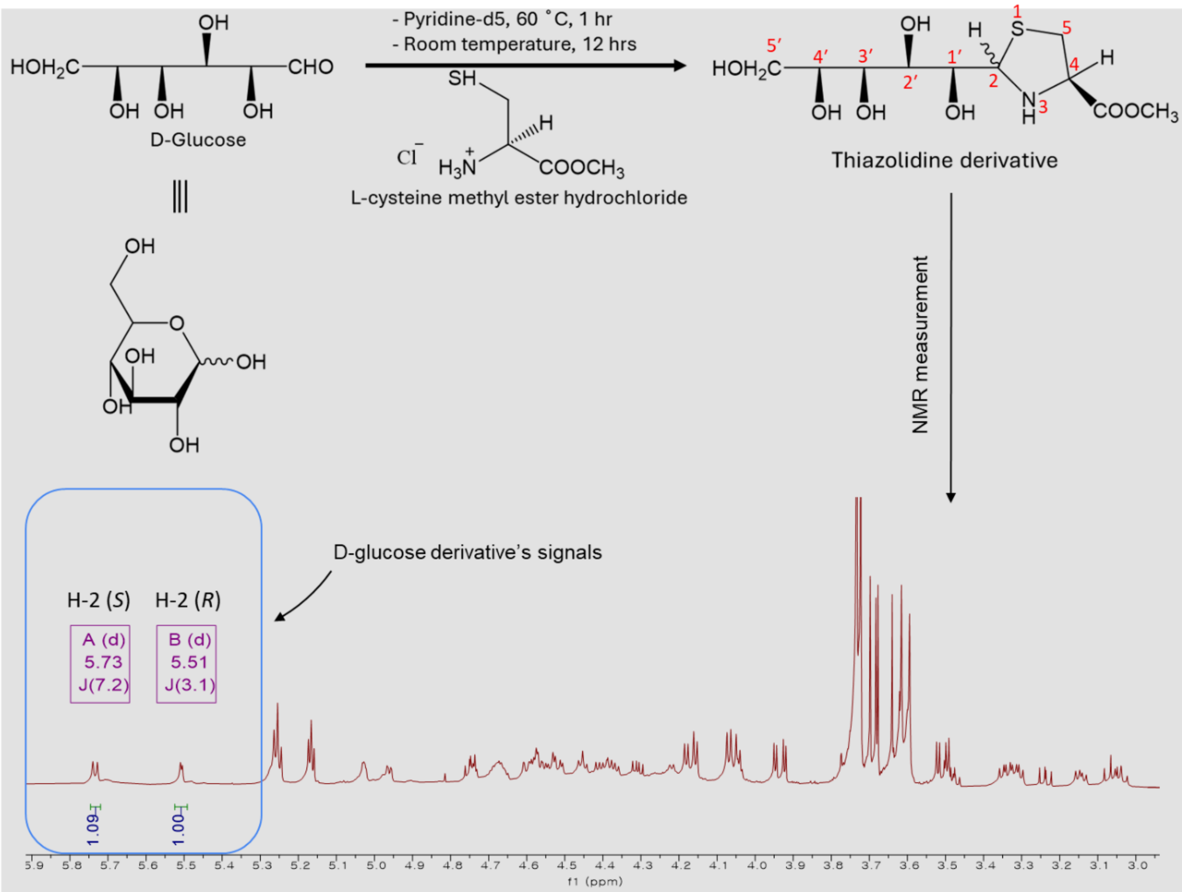


Figure S3. a, Hydrolysis of the Kudzu polysaccharide to obtain only monosaccharides. **b,** Confirmation of monosaccharide as D-glucose.

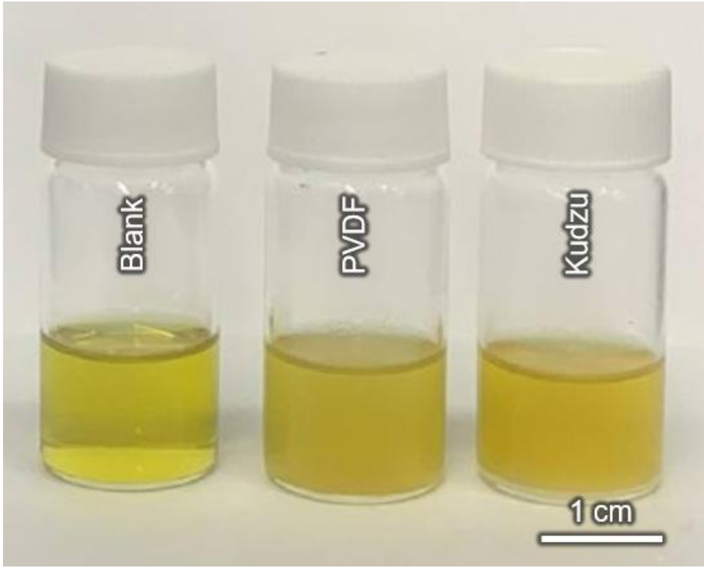


Figure S4. Digital photo of the Li_2S_6 solutions right after mixing.

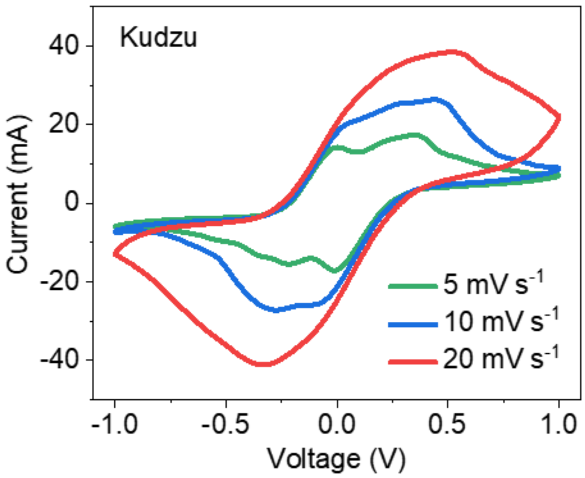


Figure S5. CV curves of a symmetric cell using the Kudzu binder at various scan rates.

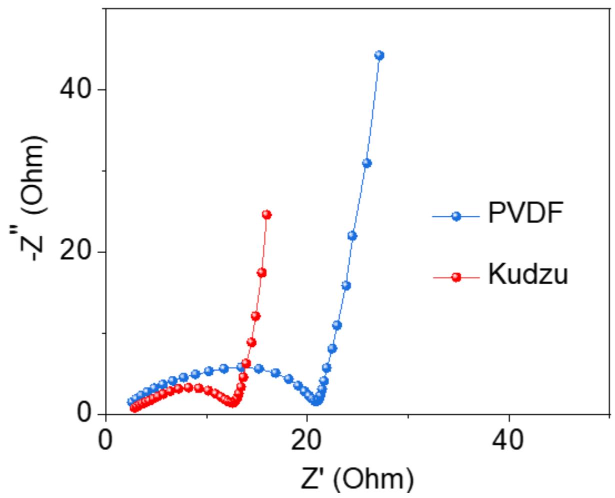


Figure S6. EIS of the symmetric cells.

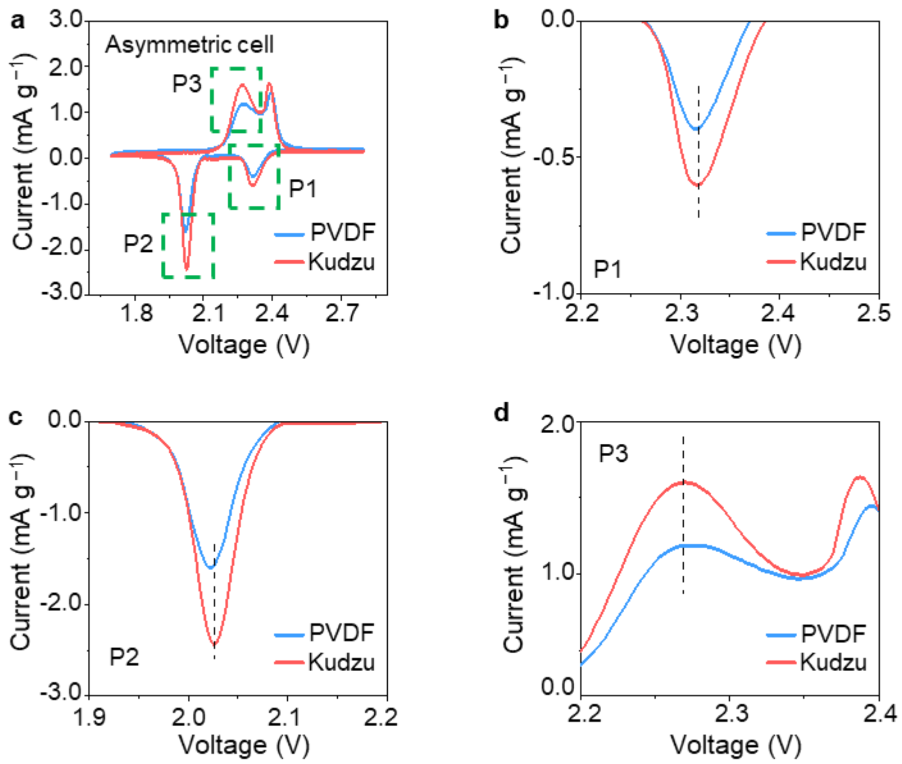


Figure S7. CV curves of the (**a**) Li–S cells using the PVDF and Kudzu binders and (**b–d**) enlarged figures corresponding to the marked areas.

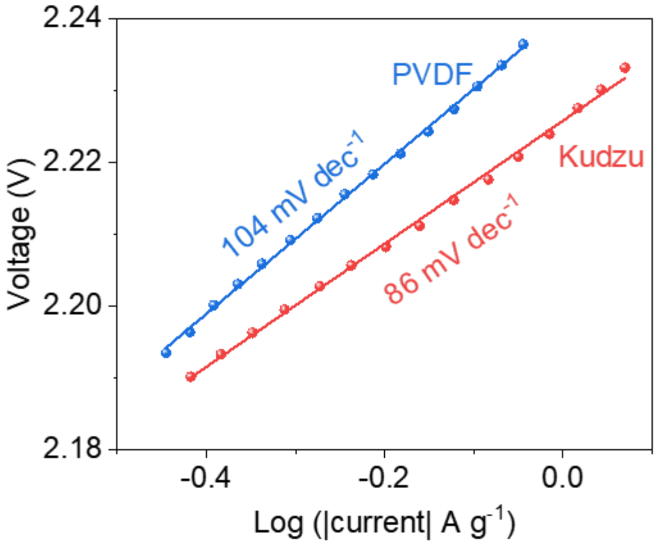


Figure S8. Tafel plots calculated from peak II (oxidation peak).

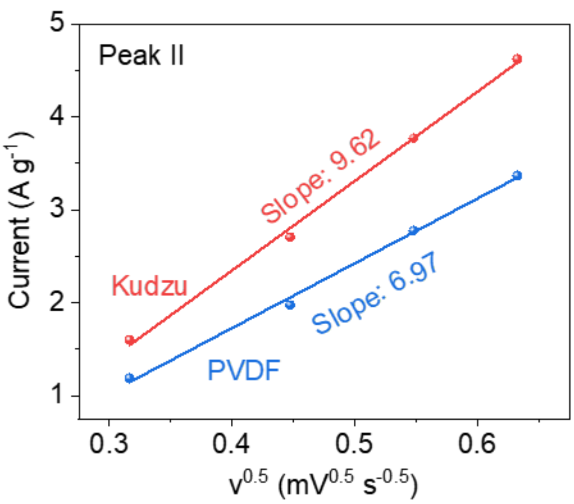


Figure S9. Linear fitting of peak current versus the square root of the scan rates for peak II.

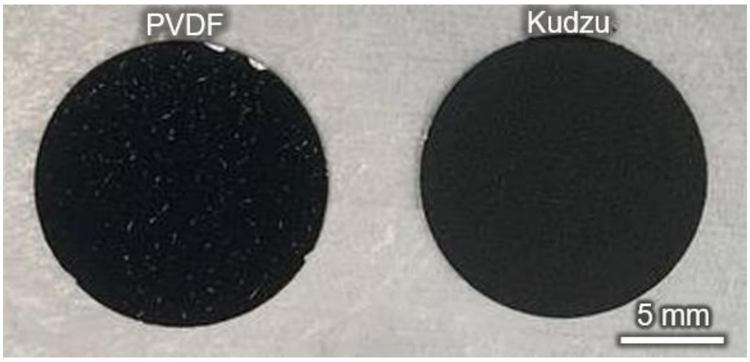


Figure S10. Digital photo of the S cathodes using the PVDF or Kudzu binder.

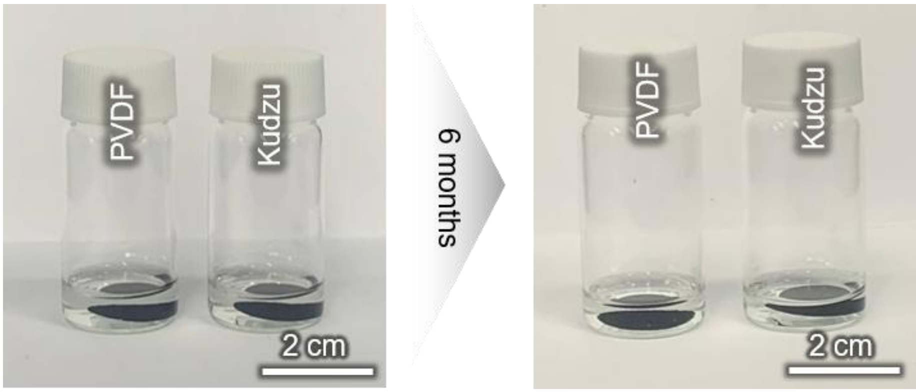


Figure S11. Digital photos of electrodes immersed in electrolyte for 6 months.

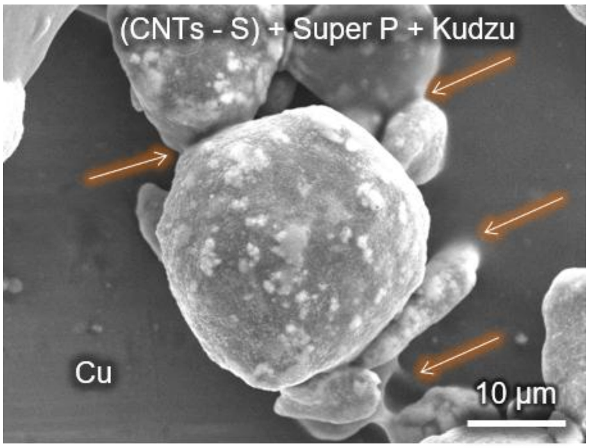


Figure S12. SEM images showing the interconnection network between the composites and between the composites and a Cu current collector.

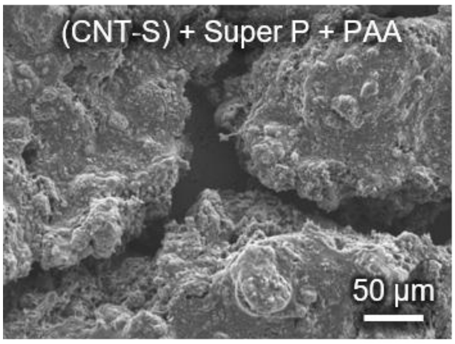


Figure S13. SEM image of the S cathode using the PAA binder.

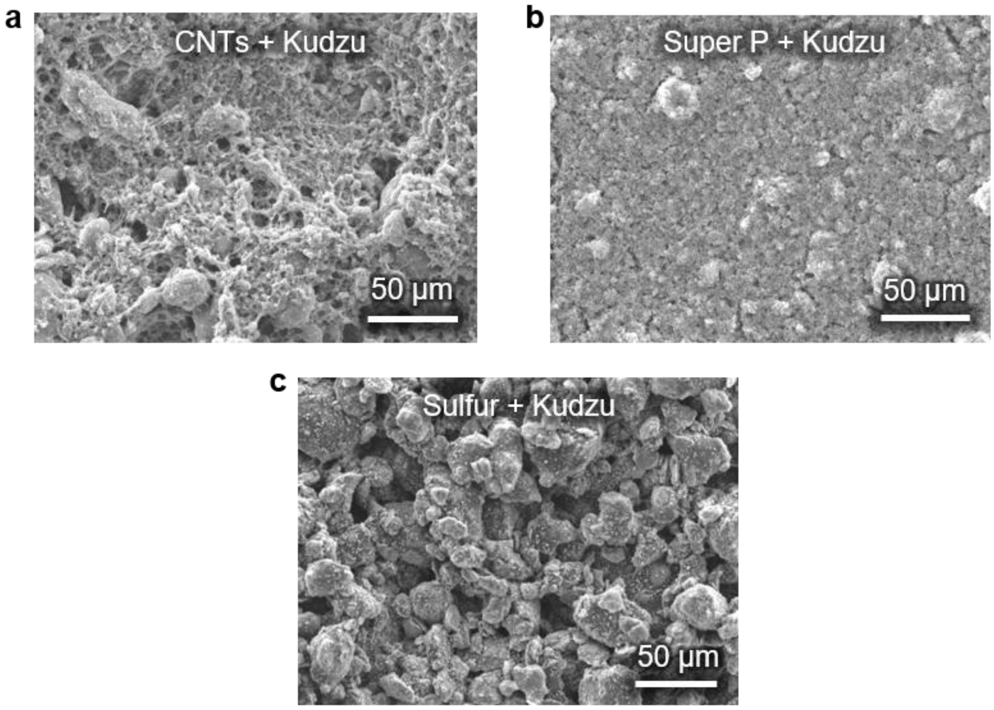


Figure S14. SEM images showing the morphologies of the (**a**) CNTs/Kudzu, (**b**) super P/Kudzu, and (**c**) S/kudzu composites.

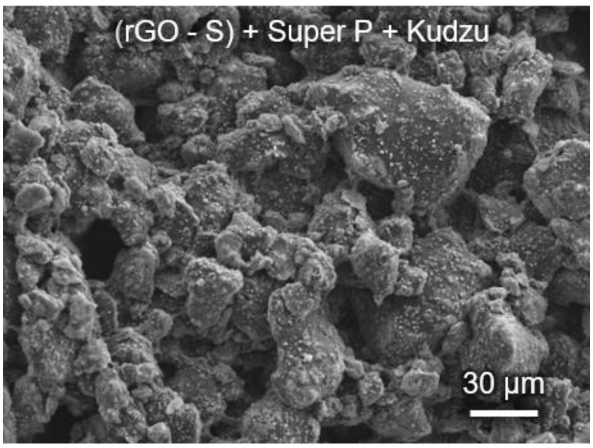


Figure S15. SEM image of the S cathode using rGO as an S host and the Kudzu binder.

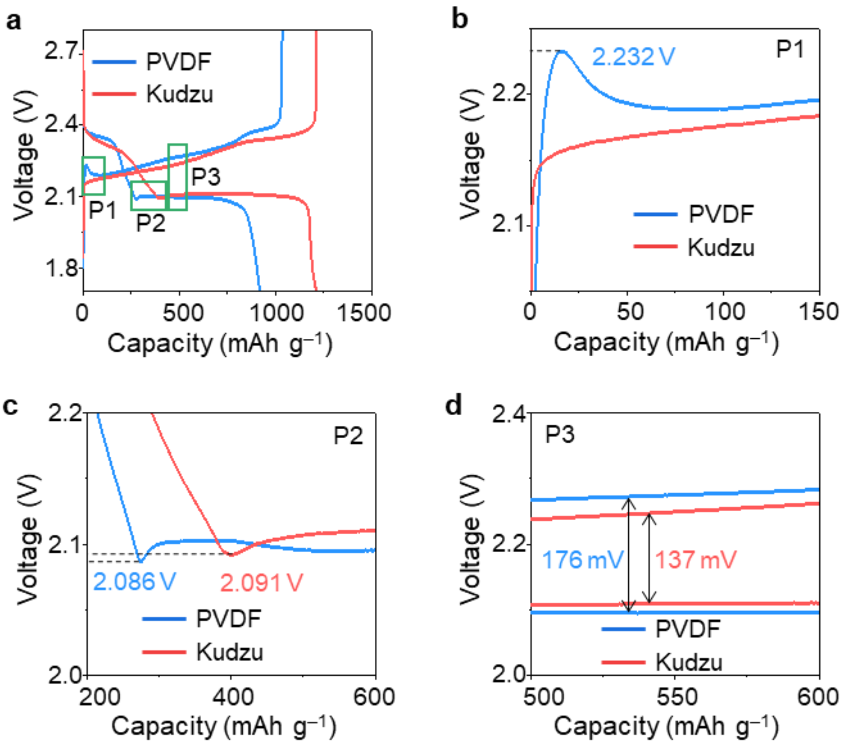


Figure S16. GCD curves of the (**a**) Li–S cells using the PVDF and Kudzu binders and (**b–d**) enlarged figures corresponding to the marked areas.

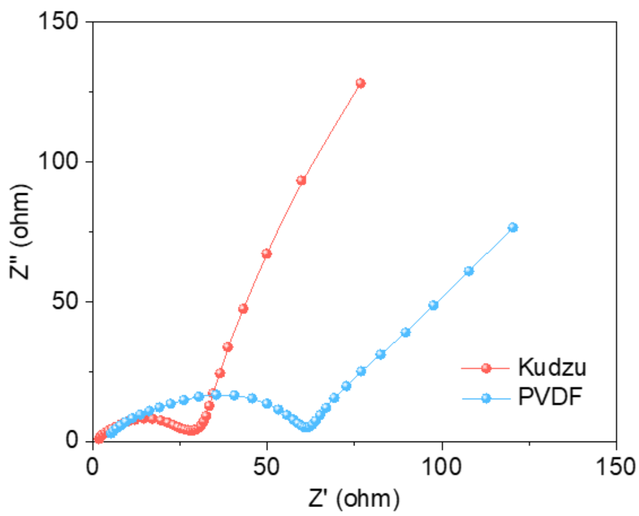


Figure S17. EIS of the Li–S cells after the first cycle.

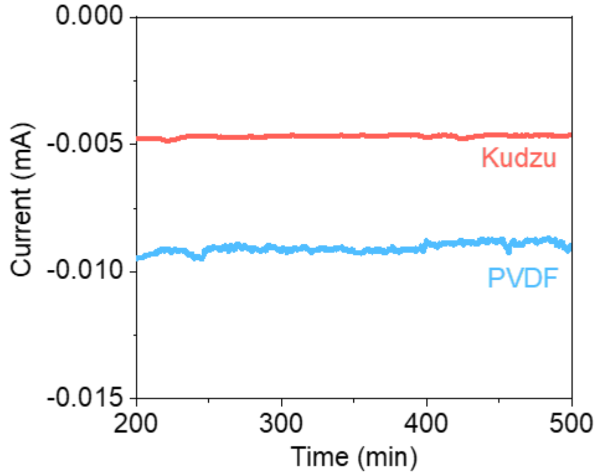


Figure S18. Shuttle currents of the Li–S cells measured by the potentiostatic mode after the cells were discharged to 2.38 V at 0.1 C

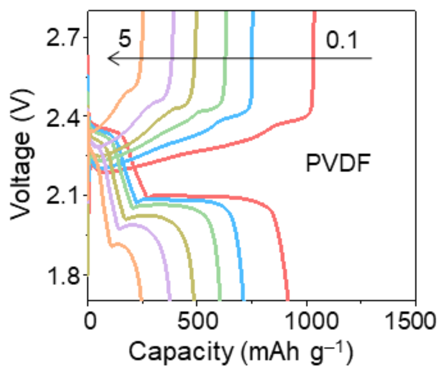


Figure S19. GCD curves of the Li–S cell using the PVDF binder at various current densities.

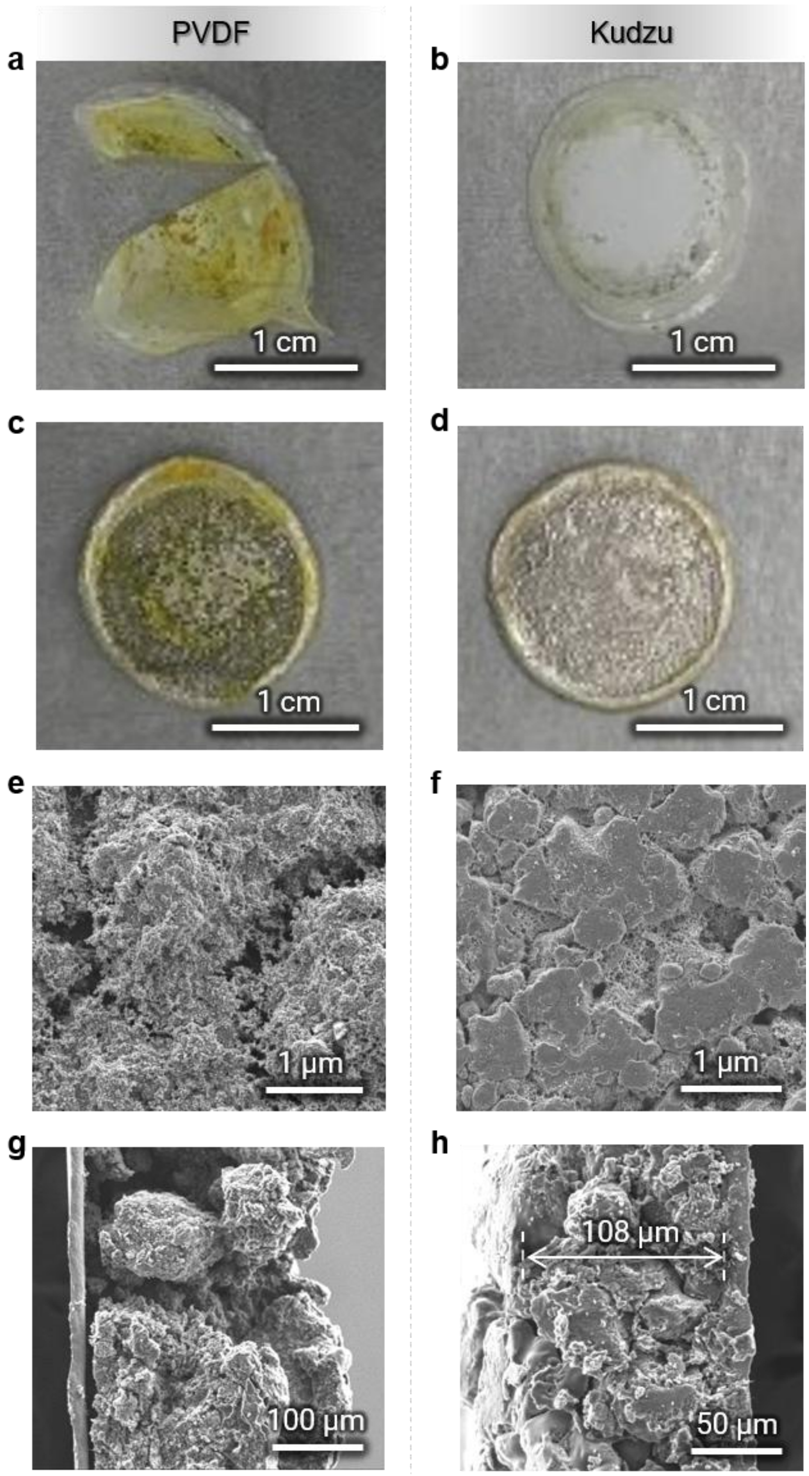


Figure S20. Post-mortem characterization of the cycled cells. **a,b,** Digital photos of the separators. **c,d,** Digital photos of the Li anodes. **e,f,** Top-view SEM images showing the morphologies of the cycled S cathodes. **g,h,** Cross-section SEM images of the cycled S cathodes.

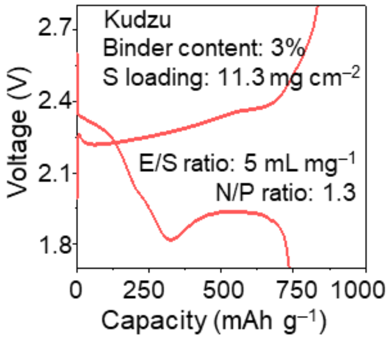


Figure S21. Representative charge–discharge curves of the cell with extremely high S loading and low Kudzu binder content.

4. Supporting Tables

Table S1. Parameters of the practical cell for gravimetric and volumetric energy density calculations.

Content	Cathode	Anode	Electrolyte	Separator	Total
Areal weight (mg cm^{-2})	11.09	2.67	35.5	1.08	50.34
Thickness (μm)	91	50	-	25	166

Note: The thickness of the electrolyte is not considered because of the infiltration of the electrolyte into the cathode and separator.

Table S2. Cycling performance comparison between the Li-S battery cell using the Kudzu binder and previously reported cells in the literature.

Binder	S loading (mg cm^{-2})	E/S ratio (μl mg^{-1})	Binder content (%)	Current density (C)	Cycle number	Capacity retention (%)	Ref.
Poly(ethylene glycol) diglycidyl ether - polyethylenimine	1.2–1.5	-	10	1.5	400	72	S2
Citric acid - betaine	1.4–1.5	-	10	1	700	66.2	S3
Soft acidic–hard basic ionomers	2	7	10	1	300	71.5	S4
Gallium-tin liquid metal - caffeic acid modified chitosan	0.9–1.1	12	10	1	500	64	S5
LiPAA	2.5	15	5	0.5	200	73.2	S6
Curdiea racovitzae-derived polysaccharide	1.3	8	10	0.5	200	50	S7
Hexamethylolmelamine - polyacrylic acid	1.2	8	10	1	300	56	S8
PAA-b-(PnBA-co-PHFBA)-b-PAA	1.5	-	10	0.5	150	64.1	S9
Kudzu	**1.5**	**10**	**10**	**1.0**	**500**	**75**	**This work**
	7.1	**5.0**	**10**	**0.1**	**100**	**73**	
	11.3	**5.0**	**3**	**0.1**	**15**	**92**	

5. Supporting References

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